Lipkin model, Entanglement, Noise, Dephasing and dissipation

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I. INTRODUCTION

Structure of article

1. Introduction with motivation
2. Discussion of the Lipkin model
3. Setting up time evolution
4. PMMs and time evolution
5.

II. MATHEMATICAL FRAMEWORK AND ENTANGLEMENT MEASURES

A. Hamiltonian and Symmetry.

The Lipkin–Meshkov–Glick (LMG) model describes N spin-$\frac{1}{2}$ particles interacting through collective spin operators. For four fermions, the case we consider in the examples here, each levels has degeneration $d = 4$, leading to different total spin values. The two levels have quantum numbers $\sigma = \pm 1$, with the upper level having $2\sigma = +1$ and energy $\varepsilon_1 = \varepsilon/2$. The lower level has $2\sigma = -1$ and energy $\varepsilon_2 = -\varepsilon/2$. That is, the lowest single-particle level has negative spin projection (or spin down), while the upper level has spin up. In addition, the substates of each level are characterized by the quantum numbers $p = 1, 2, 3, 4$.

The Hamiltonian of the system is given by

$$\begin{aligned}
\hat{H} =& \quad \hat{H}_0 + \hat{H}_1 + \hat{H}_2 \\
\hat{H}_0 =& \quad \tfrac{1}{2}\varepsilon \textstyle\sum_{\sigma,p} \sigma a^\dagger_{\sigma,p} a_{\sigma,p} \\
\hat{H}_1 =& \quad \tfrac{1}{2}V \textstyle\sum_{\sigma,p,p'} a^\dagger_{\sigma,p} a^\dagger_{\sigma,p'} a_{-\sigma,p'} a_{-\sigma,p} \\
\hat{H}_2 =& \quad \tfrac{1}{2}W \textstyle\sum_{\sigma,p,p'} a^\dagger_{\sigma,p} a^\dagger_{-\sigma,p'} a_{\sigma,p'} a_{-\sigma,p}
\end{aligned}$$

where V and W are constants. The operator H_1 can move pairs of fermions while H_2 is a spin-exchange term. The latter moves a pair of fermions from a state $(p\sigma, p' - \sigma)$ to a state $(p - \sigma, p'\sigma)$.

We are going to rewrite the above Hamiltonian in terms of so-called quasispin operators

$$\begin{aligned}
\hat{J}_+ =& \quad \textstyle\sum_p a^\dagger_{p+} a_{p-} \\
\hat{J}_- =& \quad \textstyle\sum_p a^\dagger_{p-} a_{p+} \\
\hat{J}_z =& \quad \tfrac{1}{2} \textstyle\sum_{p\sigma} \sigma a^\dagger_{p\sigma} a_{p\sigma} \\
\hat{J}^2 =& \quad J_+ J_- + J_z^2 - J_z
\end{aligned}$$

and the number operator

$$\hat{N} = \sum_{p\sigma} a^\dagger_{p\sigma} a_{p\sigma}.$$

We will now rewrite the Hamiltonian in terms of the above quasi-spin operators and the number operator

$$N = \sum_{p,\sigma} a^{\dagger}_{p\sigma} a_{p\sigma} .labeleq:N \tag{1}$$

Going through each term of the Hamiltonian and using the expressions for the quasi-spin operators we obtain

$$H_0 = \varepsilon J_z, \tag{2}$$

$$H_1 = \frac{1}{2} V \left(J_+^2 + J_-^2 \right), \tag{3}$$

and

$$H_2 = \frac{1}{2} W \left(-N + J_+ J_- + J_- J_+ \right). \tag{4}$$

The above expressions can in turn be used to show that the Hamiltonian commutes with the various quasi-spin operators. This leads to quantum numbers which are conserved. Let us first show that $[H, J^2] = 0$, which means that J is a so-called *good* quantum number and that the total spin is a conserved quantum number.

The matrix elements are given by $\langle J, J_z | H | J', J'_z \rangle$. The Hamiltonian is hermitian and we obtain after all this labor of ours

$$H_{J=2} = \begin{bmatrix} -2\varepsilon & 0 & \sqrt{6}V & 0 & 0 \\ 0 & -\varepsilon + 3W & 0 & 3V & 0 \\ \sqrt{6}V & 0 & 4W & 0 & \sqrt{6}V \\ 0 & 3V & 0 & \varepsilon + 3W & 0 \\ 0 & 0 & \sqrt{6}V & 0 & 2\varepsilon \end{bmatrix} labeleq:HJ=2 \tag{5}$$

Its Hamiltonian can be expressed as

$$\hat{H}_{\mathrm{LMG}} = -\frac{\lambda}{N} \left(\hat{S}_x^2 + \gamma \, \hat{S}_y^2 \right) - h \, \hat{S}_z, \tag{6}$$

where $\hat{S}_\alpha = \frac{1}{2} \sum_{i=1}^{N} \hat{\sigma}_i^\alpha$ are collective spin operators, λ is the interaction strength, γ the anisotropy parameter, and h an external magnetic field. The factor $1/N$ ensures a well-defined thermodynamic limit. The total spin $\hat{\mathbf{S}}^2$ commutes with the Hamiltonian, implying conservation of total spin $S = N/2$, and the system therefore lives in the symmetric subspace spanned by the Dicke states

$$|S, M\rangle \equiv |N/2, M\rangle, \quad M = -S, \ldots, S. \tag{7}$$

B. Mean-Field and Quantum Phase Transition.

In the thermodynamic limit ($N \to \infty$), a mean-field treatment becomes exact. A semiclassical energy functional can be written in terms of collective spin variables (θ, ϕ) on the Bloch sphere as

$$E(\theta, \phi) = -\frac{\lambda}{2} \sin^2\theta \left(\cos^2\phi + \gamma \sin^2\phi \right) - h \cos\theta. \tag{8}$$

The ground state undergoes a second-order quantum phase transition at $h_c = \lambda(1+\gamma)/2$, separating a symmetric phase ($\langle S_x \rangle = 0$) from a symmetry-broken phase ($\langle S_x \rangle \neq 0$). Near this critical point, the entanglement entropy exhibits logarithmic scaling with system size, signalling a diverging correlation length in the collective degrees of freedom.

C. Reduced Density Matrices and Entropy.

Because of full permutation symmetry, the reduced density matrix of any single spin is identical and can be obtained from the collective spin density operator ρ as

$$\rho_1 = \mathrm{Tr}_{2,\ldots,N}(\rho) = \frac{1}{2} \left(\mathbb{I} + \frac{2}{N} \langle \hat{\mathbf{S}} \rangle \cdot \boldsymbol{\sigma} \right). \tag{9}$$

The single-spin entanglement (or *one-tangle*) with the remainder of the system is quantified by the linear entropy

$$\tau_1 = 4 \det(\rho_1) = 1 - \frac{4}{N^2} \langle \hat{\mathbf{S}} \rangle^2. \tag{10}$$

This quantity vanishes in product states and reaches unity for maximally mixed single-spin states.

For bipartite entanglement between two subsystems of sizes L and $N-L$, the von Neumann entropy

$$S_L = -\mathrm{Tr} \left(\rho_L \log_2 \rho_L \right) \tag{11}$$

is the standard measure. In the symmetric Dicke basis, ρ_L is block-diagonal and its eigenvalues can be expressed analytically in terms of Clebsch–Gordan coefficients, allowing explicit evaluation of S_L for moderate N.

D. Two-Spin Correlations and Concurrence.

Pairwise entanglement between two spins is characterized by the *concurrence* C, defined from the two-spin reduced density matrix ρ_{12} as

$$C = \max\left\{0, \lambda_1 - \lambda_2 - \lambda_3 - \lambda_4 \right\}, \tag{12}$$

where λ_i are the square roots of the eigenvalues (in descending order) of the matrix $\rho_{12}\tilde{\rho}_{12}$, with $\tilde{\rho}_{12} = (\sigma_y \otimes \sigma_y) \, \rho_{12}^* \, (\sigma_y \otimes \sigma_y)$. In the LMG model, due to full permutation symmetry, C is identical for all spin pairs. It reaches a maximum near the critical point and decays as $N^{-1/3}$ at criticality, indicating that bipartite entanglement is sub-extensive even though the total multipartite entanglement diverges.

E. Multipartite and Mixed-State Measures.

Beyond bipartite partitions, the *global entanglement* and the *geometric measure of entanglement* are often used:

$$E_{\mathrm{G}} = 1 - \max_{|\phi_{\mathrm{sep}}\rangle} |\langle \phi_{\mathrm{sep}} | \psi_0 \rangle|^2, \tag{13}$$

where $|\phi_{\text{sep}}\rangle$ is the closest separable state to the ground state $|\psi_0\rangle$. For mixed or non-complementary partitions, the logarithmic negativity

$$\mathcal{N} = \log_2 \|\rho^{T_A}\|_1 \tag{14}$$

quantifies entanglement by means of the trace norm of the partially transposed density matrix ρ^{T_A}.

F. Scaling and Universality.

Near the quantum critical point, different entanglement measures show universal scaling with system size N:

$$S_L \sim \frac{1}{6}\log_2 N + \text{const}, \qquad \tau_1 \sim N^{-2/3}, \qquad C \sim N^{-1/3}. \tag{15}$$

These exponents coincide with those predicted by conformal field theory for one-dimensional critical systems, even though the LMG model is infinite-range. Such universality underscores that entanglement, rather than spatial locality, captures the essential signatures of quantum criticality.

III. RESULTS AND DISCUSSIONS

IV. CONCLUSION